

Advancements in Dense Plasma Focus (DPF) for Space Propulsion

Robert Thomas¹, Yang Yang², G.H. Miley³, F.B. Mead⁴

¹*Department of Aerospace Engineering, University of Illinois at Urbana – Champaign,
Urbana, Illinois 61801, USA*

^{2,3}*Department of Nuclear, Plasma, and Radiological Engineering, University of Illinois at Urbana– Champaign,
Urbana, Illinois 61801, USA*

⁴*AFRL/PRSP, 10 E. Saturn Blvd., Edwards AFB CA 93524-7680*

¹(217) 244-5348, rethomas@uiuc.edu

Abstract. The development of a dense plasma focus (DPF) propulsion device using $p\text{-}^{11}\text{B}$ is described. A propulsion system of this type is attractive because of its high thrust-to-weight ratio capabilities at high specific impulses. From a fuel standpoint, $p\text{-}^{11}\text{B}$ is advantageous because of the aneutronic nature of the reaction, which is favorable for the production of thrust since the charged particles can be channeled by a magnetic field. Different fusion mechanisms are investigated and their implication to the $p\text{-}^{11}\text{B}$ reaction is explored. Three main requirements must be satisfied to reach breakeven for DPF fusion: a high Ti/Te ratio (~ 20), an order of magnitude higher pinch lifetime, and the reflection and absorption of at least 50% radiation. Moreover, a power re-circulation method with high efficiency must be available for the relatively low Q value of the DPF fusion reactor. A possible direct energy conversion scheme using magnetic field compression is discussed. DPF parameters are estimated for thrust levels of 1000 kN and 500 kN, and possible propulsion applications are discussed, along with developmental issues.

INTRODUCTION

Filipov and Mather initially engineered the dense plasma focus (DPF) in separate endeavors. During the 1960's the DPF was first investigated for use as a power reactor, but was ultimately set aside in favor of other concepts. Since then, the DPF has been used predominantly as a laboratory source of both x-rays and neutrons. The United States Air Force (USAF) is currently investigating the DPF for its potential use as a fusion rocket propulsion system burning advanced fuels. In both the Filipov and Mather geometries, the energy stored in a capacitor bank is discharged into a coaxial set of cylindrical electrodes housed in a chamber kept at a pressure of a few torr of gas mixture. The discharge is initiated along an insulator placed at the base of the electrodes, and the rising magnetic pressure in the space moves the current sheath formed in the discharge forward between the two electrodes. Part of the fill gas is ionized and compressed at the top of the center electrode to high temperatures and densities (pinch), as shown in Figure 1. The objective of this paper is to estimate the parameters of a DPF rocket and identify the critical research areas for its development.

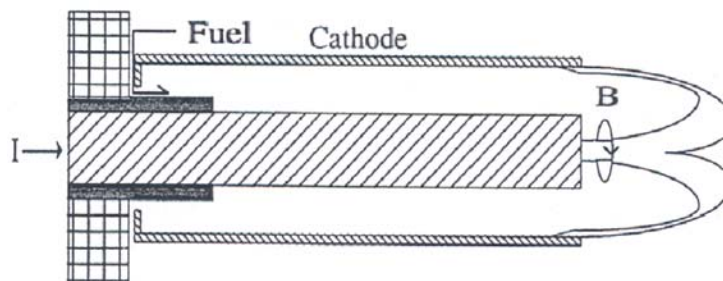


FIGURE 1. A Typical Plasma Focus.

Report Documentation Page			Form Approved OMB No. 0704-0188		
Public reporting burden for the collection of information is estimated to average 1 hour per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden, to Washington Headquarters Services, Directorate for Information Operations and Reports, 1215 Jefferson Davis Highway, Suite 1204, Arlington VA 22202-4302. Respondents should be aware that notwithstanding any other provision of law, no person shall be subject to a penalty for failing to comply with a collection of information if it does not display a currently valid OMB control number.					
1. REPORT DATE NOV 2004		2. REPORT TYPE		3. DATES COVERED -	
4. TITLE AND SUBTITLE Advancements in Dense Plasma Focus (DPF) for Space Propulsion			5a. CONTRACT NUMBER		
			5b. GRANT NUMBER		
			5c. PROGRAM ELEMENT NUMBER		
6. AUTHOR(S) Robert Thomas; Yang Yang; G Miley; F Mead, Jr			5d. PROJECT NUMBER 4847		
			5e. TASK NUMBER 0159		
			5f. WORK UNIT NUMBER		
7. PERFORMING ORGANIZATION NAME(S) AND ADDRESS(ES) Air Force Research Laboratory (AFMC),AFRL/PRSP,10 E. Saturn Blvd.,Edwards AFB,CA,93524-7680			8. PERFORMING ORGANIZATION REPORT NUMBER		
9. SPONSORING/MONITORING AGENCY NAME(S) AND ADDRESS(ES)			10. SPONSOR/MONITOR'S ACRONYM(S)		
			11. SPONSOR/MONITOR'S REPORT NUMBER(S)		
12. DISTRIBUTION/AVAILABILITY STATEMENT Approved for public release; distribution unlimited					
13. SUPPLEMENTARY NOTES					
14. ABSTRACT The development of a dense plasma focus (DPF) propulsion device using p-11B is described. A propulsion system of this type is attractive because of its high thrust-to-weight ratio capabilities at high specific impulses. From a fuel standpoint, p-11B is advantageous because of the aneutronic nature of the reaction, which is favorable for the production of thrust since the charged particles can be channeled by a magnetic field. Different fusion mechanisms are investigated and their implication to the p-11B reaction is explored. Three main requirements must be satisfied to reach breakeven for DPF fusion: a high Ti/Te ratio (~20), an order of magnitude higher pinch lifetime, and the reflection and absorption of at least 50% radiation. Moreover, a power re-circulation method with high efficiency must be available for the relatively low Q value of the DPF fusion reactor. A possible direct energy conversion scheme using magnetic field compression is discussed. DPF parameters are estimated for thrust levels of 1000 kN and 500 kN, and possible propulsion applications are discussed, along with developmental issues.					
15. SUBJECT TERMS					
16. SECURITY CLASSIFICATION OF:			17. LIMITATION OF ABSTRACT	18. NUMBER OF PAGES 8	19a. NAME OF RESPONSIBLE PERSON
a. REPORT unclassified	b. ABSTRACT unclassified	c. THIS PAGE unclassified			

FUSION MECHANISMS

Different models have been suggested for the production of neutrons in the plasma focus. A brief investigation of these models can give insight into the mechanisms of the $p\text{-}^{11}\text{B}$ fusion reaction. The moving-boiler model suggests that the ion velocity is isotropic in the center of mass of the system, but this center of mass moves along the axis in the laboratory frame. This model gives a much higher ion temperature than the estimates based on other observations. Also, measurements have shown (Milanese, 1978) that neutron and electromagnetic radiation emissions from plasma focus devices are characterized by a marked anisotropy. The beam-target model suggests that at the collapse of the current sheath, an electric field is generated by the $m = 0$ sausage magnetohydrodynamic instability. The charged particles are accelerated by this intense electric field into high velocities and collide with stationary targets. However, the duration of neutron production, typically 100 nanoseconds, is too long for the attributed magnetohydrodynamic instability, and an unrealistically high ion current (greater than 10^6 A) must be assumed.

W.H. Bostick and V. Nardi (Bostick, 1975) studied the fine structure inside the plasma columnar region during the stage of neutron production, and on the current sheath that forms this column. The fine structure can be depicted as a texture of numerous filaments with a diameter ranging from a few millimeters down to a half a millimeter, and can be observed in all regions of the current sheath before, during, and after the axial-collapse time. Their experimental data indicates that high-energy electrons and ions up to and above the nuclear reaction threshold are concentrated in localized regions of the current sheath during the neutron production time. The acceleration of ions to high energies inside localized plasma regions with linear dimensions of a fraction of a millimeter is well explained in terms of decay of the observed magnetic structure of the current sheath. The acceleration of ions by inductive fields during the structure decay accounts for the anisotropy of neutron and x-ray energy spectrum. Based on a multiplicity of experimental observations on the bremsstrahlung x-ray emission from localized sources, Bostick and Nardi estimated the peak value of the magnetic field intensity as $B_{\text{max}} \sim 2 \times 10^8$ G ($\pm 50\%$). They also evidenced the generation of electron and ion beams by focused discharges and obtained an ion energy spectrum using a time-of-flight method (Bostick, 1978). All of the empirical evidence shows that the bulk of the fusion reactions occur in a multiplicity of localized regions with a density of $> 10^{20} \text{ cm}^{-3}$.

Due to the varieties of different fusion models, a positive or negative conclusion can be formed depending on the representation. Among the models, the moving boiler model provides the strongest opposition to $p\text{-}^{11}\text{B}$ reaction. Since the ions and electrons are in thermal equilibrium, the bremsstrahlung radiation will ruin any possibility of breakeven. Fortunately, there is substantial evidence showing non-thermal processes occur inside the plasma focus discharges. To achieve break-even and ignition for a $p\text{-}^{11}\text{B}$ reaction, the electron temperature needs to be decreased while keeping ions at high temperature. The existence of electron filaments and ion clusters, as depicted in the fine structure model, provides such a possibility. Since the bulk ions and electrons are located in different regions, the heating mechanism can be greatly depressed. Therefore a much lower radiation level and electron temperature can be achieved.

PARAMETER ESTIMATION

A cylindrical plasma, whose dimensions remain constant during the stable pinch lifetime, approximates the pinch. To simplify the DPF pinch physics, it is assumed that the pinch is stable for the time it takes the capacitors to discharge their energy at the maximum current and voltage (dc). Theoretically, this estimate of pinch lifetime is a best-case scenario. The maximum current flows through the pinch, and the pinch is stable for as long as the capacitor banks are charged. Thus any subsequent calculations of energy yield from this pinch will represent an upper limit. However, when comparing empirical scalings to high bank energies, the dc assumption gives a conservative pinch lifetime prediction. During the rundown phase, a LC circuit approximates the DPF. The resulting circuit equations can be solved in dimensionless form to obtain the voltage and current profiles. As mentioned earlier, there is a strong possibility of obtaining an electron temperature that is far less than the ion temperature in the DPF device. The Q value as a function of T_i/T_e is shown in Figure 2. For DPF operation, it is also essential that part of the bremsstrahlung radiation is reflected back into the core. Figure 3 shows the bremsstrahlung losses and fusion heating power as a function of the electron temperature assuming a 60% reflection rate.

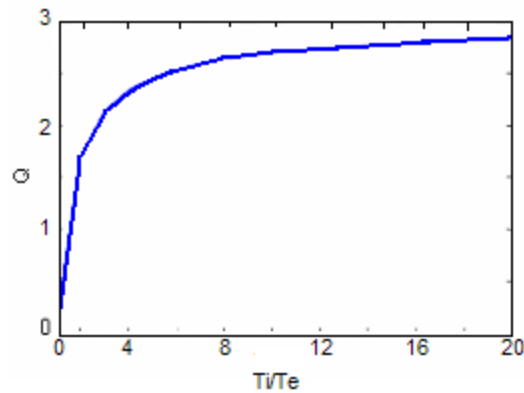


FIGURE 2. Q Value as a Function of T_i/T_e .

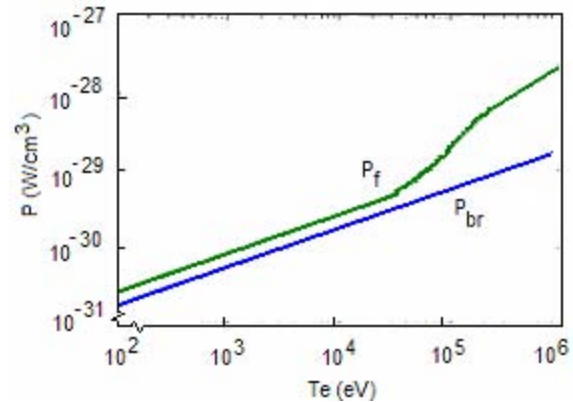


FIGURE 3. Bremsstrahlung Losses and Fusion Heating Power vs. T_e .

Parameter Optimization

To get a maximum Q value, the geometric dimensions of the DPF electrodes must be optimized. Figure 4 shows Q value as a function of electrode radius for the case of a capacitor bank energy of 80 MJ. Using the optimized electrode diameters, the Q value as a function of initial particle density is shown in Figure 5. This figure is used as a criterion for the selection of initial particle density.

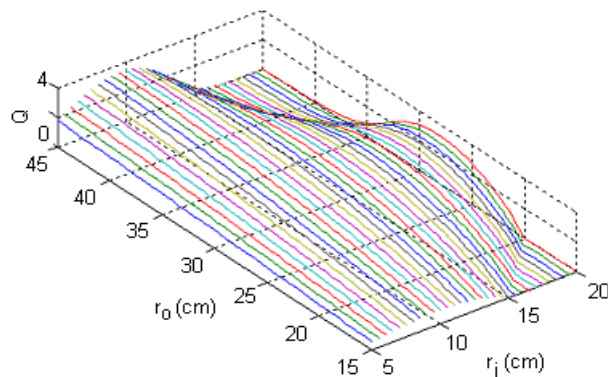


FIGURE 4. Q Value as a Function of Electrode Radius.

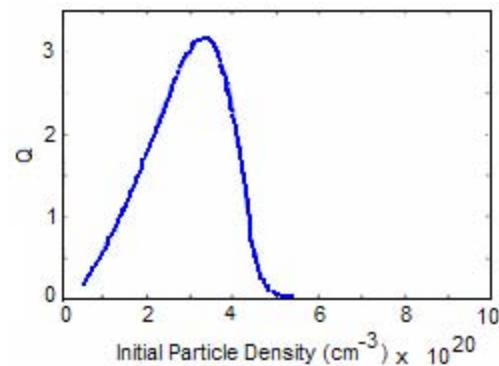


FIGURE 5. Q Value as a Function of Initial Particle Density.

Result Analysis

Using the above model and assumptions, codes were developed to calculate plasma temperature evolution and energy yield. The overall pinch lifetime was divided into small time intervals, and in each interval the plasma temperature is assumed to be a constant. The Runge-Kutta method was used to estimate the time rate of change of the plasma energy. Starting with a capacitor bank energy of 80 MJ, an initial rate of current change of 1 MA/ μ s, and a rate of inductance change of 20 m Ω (1 H/s = 1 Ω), the fusion and bremsstrahlung energy yields are 248 MJ and 38 MJ, respectively. The resulting Q value is 3.07. Firing at a repetition rate of 10 Hz, the thermal power is the sum of the input (capacitor) and output (fusion, radiation) powers, and has a value of 3674 MW.

The propellant flow required to absorb the thermal power depends on the temperature increase the propellant can handle, which depends on material temperature limitations. We assume the propellant starts at 50 K, and the walls limit the

propellant high temperature to 3000 K. This corresponds to an enthalpy difference of 105 MJ/kg, which can be used to find a mass flow rate of $\dot{m}_{prop} = 35$ kg/s. The thrust and Isp are then found to be:

$$F = \dot{m}_{prop} v_{ex} = 449 \text{ kN},$$

$$I_{sp} = \frac{v_{ex}}{g} = 1311 \text{ s.} \quad (1)$$

The required pulse power, energy, and voltage are 800 MW, 80 MJ, and 400 kV. The estimated DPF mass is 16 tons. An increase in Isp will heighten the payload capacity of a mission, making an increase in Isp desirable. For instance, to increase the specific impulse to 2000 s, the mass propellant mass flow rate would need to be decreased by 55% and the bank energy increased to 120 MJ. The corresponding DPF mass is approximated to be 24 tons. Similarly, for a 1000 kN thrust level, with an Isp of 2000 s, the DPF mass would be 48 tons. Lower thrust levels and higher specific impulses can be obtained by varying the amount of propellant used. By expelling the fusion products for thrust while exhausting only the hydrogen necessary to cool the thruster and mixing chamber walls, a thrust of 477 N and an Isp of 1.22×10^6 s is found. Due to the high mass of the capacitor banks, the thrust-to-weight ratio of such a configuration would be too small to be useful. An optimal balance between the thrust and Isp depends on the nature of the mission. An advantage of using a pickup coil as opposed to a turbine-based method for direct energy conversion system is that a lot of the material limitations are eliminated. The use of a magnetic nozzle, which limits the contact between the plasma exhaust and the nozzle material also allows for higher specific impulses. Figure 6 shows a schematic of the thruster system.

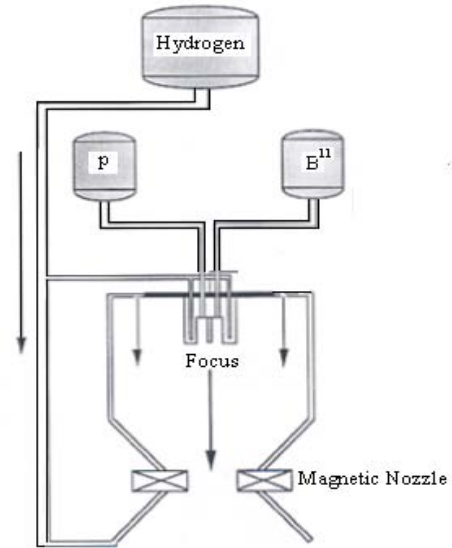


FIGURE 6. Thruster System Schematic.

Mass Estimation

An important factor for judging the performance of the DPF propulsion system is the total system mass. Although DPF fusion devices themselves would be relatively small and lightweight, they would have to be powered by electrical subsystems that could generate 10-100 pulses of multi-megajoule energy. If this mass is too high, thrust-weight ratios are reduced, thereby cutting performance (Choi, 1992).

Capacitor mass is a dominant component of the overall system mass, and any reduction of this is highly desirable. The development of new solid and liquid materials, in conjunction with advanced methods of manufacturing technology, is feasible with tools emerging from present successful technology programs. The advancement of capacitor technology to date has been successful through the capacitor developers in industry integrating new materials into advanced capacitor designs. Figure 7 (Sarjeant, 1998) shows the timeline of increasing energy density in energy discharge capacitors.

Metallized technology capacitors from US industry have been in the field for several years, at energy densities of 0.5 – 1.5 kJ/kg. Projections of energy densities beyond these levels will only be as accurate as fundamental research in dielectrics moves into the arena where energy density is indeed one of the material parameters investigated. From Figure 7, we can see a practical limit for the capacitor bank specific mass is 15-20 kJ/kg. By assuming an order of magnitude increase in specific mass from what is available now (~0.4 kJ/kg) and assuming the capacitor mass to be half the DPF system mass, Figure 8 is obtained. The thrust-to-weight increases linearly as the capacitor bank energy ranges from 20 MJ to 100 MJ. However,

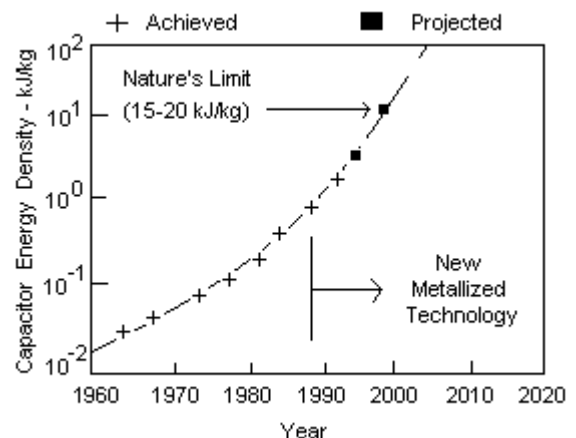


FIGURE 7. Projected Capacitor Energy Density.

as the capacitor bank energy exceeds 100 MJ, the capacitor bank becomes dominant and the thrust-to-weight ratio levels off. One can also see that the thrust-to-weight ratio is heavily dependant on the DPF thruster efficiency. The thruster efficiency is defined as the jet power divided by the DPF power input.

The technical factors which are critical in the specification and design of pulse discharge capacitors are energy density, required lifetime, pulse shape and average power density. Dielectric materials are key to advanced capacitor technology for the future. Higher dielectric strength and improved resistance to aging under electrical stress will lead to a higher energy density and a longer lifetime. If the dielectric constant can be increased without a commensurate increase in dissipation, higher energy and power densities can be achieved. Energy density of film capacitors has historically been low, but recent developments at Maxwell have increased their energy density by an order of magnitude, and further advances are anticipated (Ennis, 1990). For missions requiring long-term permanent presence such as colonies, the capacitor technology employed must not only possess minimum mass and volume but also the ultimate in reliability and stable operating characteristics.

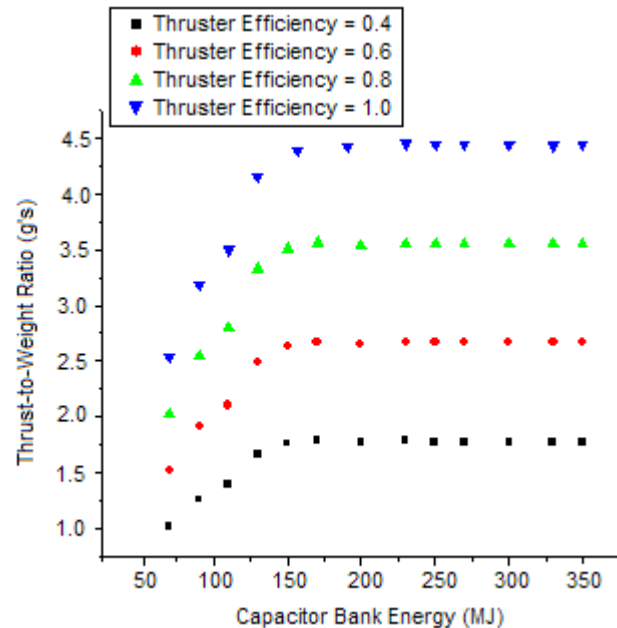


FIGURE 8. Thrust-to-Weight Ratio vs. Bank Energy.

DIRECT ENERGY CONVERSION

The question may be raised as to the best, most efficient way of utilizing the energy liberated for electric power generation. The conventional turbine-based methods of producing electricity (McKee, 1992), which involve moving parts of heavy cumbersome machinery, may not in this case be the most practical and efficient way of electric generation (Neuringer, 1960). A possible method of electric generation that does not involve moving parts is generation by magnetic compression using the highly conductive plasma of the reaction products as the working fluid. A diagram of such a system incorporated into the DPF system is shown in Figure 9.

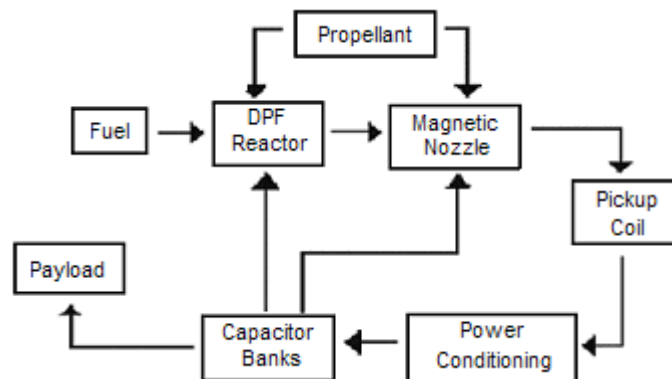


FIGURE 9. DPF Power Balance.

The principle of operation is as follows: (1) Nozzle expansion of the plasma exhaust, (2) magnetic flux trapping between the plasma jet and the stator pick-up coil with magnetic flux diffusion into both, (3) rapid field compression as the kinetic energy of the plasma is transformed into magnetic pressure, and (4) magnetic field lines forced out of the

stator ring, inducing a voltage pulse which is sent to appropriate electrical transduction equipment. A diagram of such a system is illustrated in Figure 10 (Note: Much of following was inspired by Litchford, 2001).

In systems of this type, magnetic flux is trapped between the fast moving plasma and the stationary stator because of circulating currents that are induced within the field compression surfaces. The magnetic flux can be trapped only if the plasma stream and stator have electrical conductivities that are high enough to resist magnetic field penetration during the compression process. In fact, magnetic diffusion losses can have an immense effect on system performance. Therefore, if one hopes to achieve good system performance, it is essential that diffusion losses be minimized. This requires that both the stator and plasma possess low magnetic diffusivity properties.

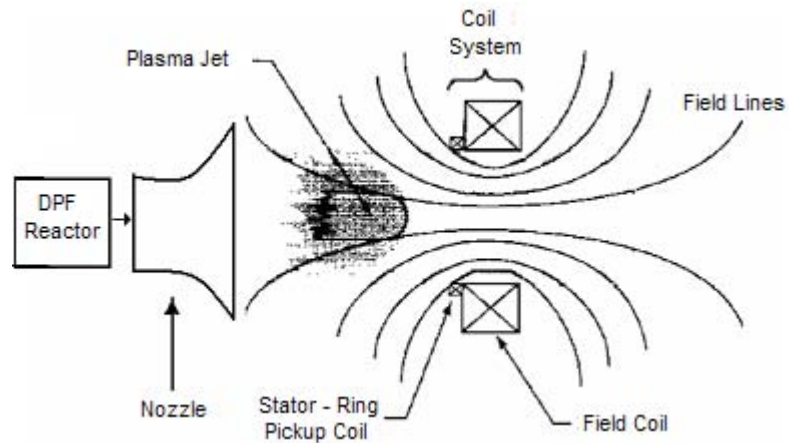


FIGURE 10. Diagram of a Magnetic Pickup Coil System.

Magnetic field compression calculations can be based on the equation for the voltage induced in a solenoid by a changing magnetic field (Conger 1967):

$$V = N(d\phi/dt) \times 10^{-8}. \quad (2)$$

From the viewpoint of global energy conversion, the direct energy conversion system obeys the following relation:

$$W_{jet} + W_{elec} = W_k - W_p = \eta_k W_k. \quad (3)$$

Ideally, the energy conversion efficiency can exceed 90%, but real plasma processes such as flute instabilities and electron Joule heating effects could reduce this value substantially. Furthermore, the magnetic field configuration and propulsion chamber design play a significant role in terms of magnetic diffusion losses and magnetic flux compression efficiency. For a propulsive system, one is interested in achieving the highest possible W_{jet} while also extracting the minimum necessary electrical energy that is needed to recharge the capacitor banks.

PROPULSION APPLICATIONS

Comparison between the DPF and other existing and proposed systems show the DPF to be one of the few systems capable of combining high specific impulse with a large thrust. The thrust projected for the DPF rates it as one of the more powerful fusion propulsion systems. Four main mission classes exist: surface-to-orbit, inner solar system, outer solar system, and extra solar. Propulsion system requirements for each class differ greatly, but the different regimes can be usefully characterized by three parameters: the specific power α , the specific impulse I_{sp} , and the thrust-to-weight ratio (Teller, 1992). The DPF propulsion system can have a wide range of values for these parameters, depending on the operational mode. Three possible modes of operation for the DPF propulsion system have been identified (Leakas, 1991). Two modes involve firing the DPF thruster over a period of time comparable to the trip time, or “continuously.” Of these two modes, one would use the fusion reaction products directly to produce a high exhaust velocity, while the other would add a constant stream of propellant to the fusion products to increase the mass flow. The final mode would involve firing the thruster for a short period of time while exhausting large quantities of propellant. This “impulsive” mode increases the thrust-to-weight ratio while still maintaining a relatively high I_{sp} . This is the mode that will likely be used for space exploration, while a “continuous mode” can be used for shorter flight times, such as an earth-to-orbit (ETO) mission. An attractive way of doing this would involve combining the DPF with chemical airbreathing technology. The general concept would involve airbreathing propulsion for Single State to Orbit (SSTO) to about half orbital speed within the earth’s sensible atmosphere, and then rocket propulsion for continued acceleration until orbital

speed is reached. Calculations have shown that the airbreathing/DPF combination is competitive with other airbreathing combination technologies (Miley, 2004).

Interplanetary flights would offer still more attractive possibilities, largely because the transit times for such flights could be reduced drastically from those expected for conventional or nuclear-thermal missions. This is a direct result of the much larger specific impulse available from the DPF propulsion system than from other high thrust engine options. A comparison of the DPF with other high thrust propulsion systems is shown in Table 1. The J-2 engine's most popular use has been as an upper stage on the Saturn V rocket. The NERVA engine uses a fissile material to thermally heat a working fluid, usually hydrogen. The QED (Bussard, 1993) uses the inertial fusion products to heat a propellant, which is directed for thrust. The table values for the NERVA and QED use point design values.

TABLE 1. Comparison of High Thrust Propulsion Systems.

Engine	J-2	NERVA	QED	DPF
Rocket Type	Chemical	Nuclear Thermal	Inertial Fusion	Magnetic Fusion
Mission Class	Lunar	Manned Interplanetary	Trans-Atmospheric, Interplanetary	Trans-Atmospheric, Lunar
Fuel	LH2 / LOX	UF ₆	p- ¹¹ B	p- ¹¹ B
Thrust (kN)	890	334	110	500
Isp (s)	418	825	3500	2000
Thrust-to-weight	72	~4	~3	~4

The DPF would fly as a "high thrust" spacecraft, under force accelerations one to two orders of magnitude larger than the local acceleration of the sun's gravity field throughout the flight. For longer space missions, the DPF may need to be fired in the "impulsive" mode. If this were the case, it may be necessary to occasionally power down the reactors. This would introduce the problem of remote restart capability, which would deserve further deliberation. Nonetheless, the high Isp and thrust-to-weight ratio of the DPF make it a forerunner in space propulsion systems.

ENGINEERING ISSUES

Three requirements must be satisfied to reach breakeven for p-¹¹B DPF fusion: a high Ti/Te ratio (~20), an order of magnitude higher pinch lifetime, and the reflection and absorption of at least 50% of the radiation power. Strong magnetic fields may reduce the collisional energy transferred from ions to electrons, which would allow average electron energies to stay far below average ion energies, and thus reduce x-ray cooling. Theoretical studies have also shown (Filippov 1997) that the collisions of alpha particle products preferentially heat boron ions faster than electrons for Te > 125 keV. Pinch lifetimes, which are on the order of 150 ns, can possibly be made longer through the use of embedded magnetic fields that prevent instability modes. Since both hard and soft x-rays have been detected in experiments (Bernstein, 1969), broadband x-rays should be used. Advances in W/Si x-ray supermirrors (Joensen, 1997) show significant promise for the reflection of bremsstrahlung radiation. Due to the high electrical needs of the DPF, capacitor specific masses need to be increased by at least an order of magnitude. Higher dielectric strength and improved resistance to aging under electrical stress will lead to higher energy densities and longer lifetimes. A lightweight, efficient method of recharging the capacitor banks between pulses also needs to be developed. A relatively straightforward configuration would use magnetic field compression from the DPF fusion products. Suitable pick-up coil materials need to be found before this technique can be verified experimentally.

CONCLUSIONS

Different fusion mechanisms have been investigated and their implications to the p-¹¹B reaction have been revealed. The existence of electron filaments and ion clusters, as depicted in the fine structure model, provides a strong possibility for achieving a high Ti/Te ratio. The three main requirements that must be satisfied to reach breakeven for DPF fusion have been identified: a high Ti/Te ratio, a magnitude higher pinch lifetime, and reflection and absorption of at least 50% of the radiation power. Several strategies for decreasing the ignition power have also been presented. Assuming the three requirements for breakeven are attainable, Matlab codes were developed to calculate plasma temperature evolution and energy yield. It was found that for a 500 kN, 2000 sec specific impulse DPF system, the required capacitor bank

energy is 120 MJ. Similarly, for a 1000 kN thrust level, the required bank energy is 240 MJ. The estimated mass for each system are 24 tons and 48 tons, respectively. Capacitor bank technology and projected advances have been discussed. An order of magnitude increase in capacitor specific mass will be necessary for a practical DPF system, and such advances are feasible due to ongoing research at Maxwell Laboratories. A direct energy scheme has been investigated that will utilize a pickup coil and the reaction products of the fusion reaction. Such a system is attractive because of its simplicity and lack of moving parts. A comparison of the DPF system with other high thrust systems show that the DPF rocket could be attractive for earth-to-orbit missions, as well as fast lunar/ interplanetary missions.

ACKNOWLEDGEMENTS

This work was performed under the joint sponsorship of The Air Force Research Laboratory Edwards Air Force Base and the Fusion Studies Laboratory at the University of Illinois.

NOMENCLATURE

ϕ	= total magnetic flux (Wb)
N	= number of turns in solenoid
W_{jet}	= propulsive jet energy of the exhaust (J)
W_{elec}	= useful extracted electrical energy (J)
W_k	= initial kinetic energy of the plasma jet (J)
W_p	= energy dissipated in the plasma jet (J)
W_s	= energy dissipated in the stator circuit (J)
η_k	= global energy conversion efficiency (J)
V	= induced voltage (V)
v_{ex}	= exhaust velocity (m/s)

REFERENCES

- Bernstein, M.J., "Space, Time, and Energy Distributions of Neutrons and X Rays from a Focused Plasma Discharge," in *Phys. Fluids*, **12**, p. 2193, (1969).
- Bostick, W.H., "Production and Confinement of High-density Plasmas," in *Annals of The New York Academy of Sciences*, **251**, p. 2, (1975).
- Bostick, W.H., "Production of GW Electron and Ion Beams By Focused Discharges," in *Second International Conference on Energy Storage, Compression, and Switching*, edited by Nardi, V., Plenum Press, New York, 1978, p. 267.
- Bostick, W.H., "Observation of 10^8 Gauss Fields and Production of 14-MeV D-T Neutrons in a Deuterium Plasma," IAEA-CN-33/E 6-3, 109, (1975).
- Bussard, R., "The QED Engine System: Direct –Electric Fusion-Powered Rocket Propulsion Systems," in *Space Nuclear Power and Propulsion*, **271**, p. 1601, (1993).
- Choi, C., Nakafuji, G., "Engineering Consideration for the Self-Energizing Magnetoplasma dynamic (MPD)- Type Fusion Plasma Thruster," Annual Report, PL-TR-92-3043, 1992.
- Conger, R.L., "Large Electric Power Pulses by Explosive Magnetic Field Compression," in *Jour. App. Phys.*, **38**, p. 2275, (1967).
- Ennis, J.B., Haskell, D.K., Sevigny, J.A., "Development of Energy Storage/ Pulse Discharge Capacitors," in *Power Sources Symposium*, p. 392, (1990).
- Filippov, N.V., et al., "Megajoule scale plasma focus as efficient X-ray source," in *Phys Lett. A*, **211**, p. 168, (1997).
- Joenson, K.D., "Broad-band hard X-ray reflectors", in *Nuclear Instruments and Methods in Physics Research B*, **132**, p. 227, (1997).
- Leakas, C.L., Choi, C.K., "The Dense Plasma Focus Fusion Thruster for Space Propulsion," in *Fusion Technology*, **20**, (1991).
- Litchford, R.J., Robertson, G.A., "Magnetic Flux Compression Reactor Concepts for Spacecraft Propulsion and Power," NASA/TP-2001-210793, 2001.
- Miley, G.H., Thomas, R., Yang Y., "Advancements in Dense Plasma Focus for Space Propulsion," FSL Report EAFB 01, 2004.
- McKee, J.W., "Space Fusion Propulsion Developments Offer a Path to Central Station Fusion Power Plants," in *43rd Congress of the International Astronautical Federation*, 1992, Washington DC, IAF-92-0616.
- Milanese, M.M., "Evidence of Non-thermal Processes in a 1-MJ Plasma Focus Device by Analyzing the Neutron Spectra," in *Nuclear Fusion*, **18**, p. 4, (1978).
- Neuringer, J.C., "Optimum Power Generation From a Moving Plasma," in *Journal of Fluid Mechanics*, **7**, p. 287, (1960).
- Sarjeant, W.J., Zirnheld, J., MacDougall, F.W., "Capacitors," in *IEEE Transactions on Plasma Science*, **26**, p. 1368, (1998).
- Teller, E., Glass, A.J., Fowler, T.K., "Space Propulsion by Fusion in a Magnetic Dipole," in *Fusion Technology*, **22**, (1992).